Electronic Spreadsheet

The big idea

A **spreadsheet** is a grid of rows and columns used to store, calculate and analyse data – marks, budgets, sales, attendance. **LibreOffice Calc** is a free spreadsheet program. Its files are called **spreadsheets** (workbooks) with the extension **.ods**, and each file contains one or more **sheets** (worksheets).

The Calc interface

- **Columns** are labelled with letters (A, B, …, Z, AA, AB, …). **Rows** are numbered (1, 2, 3, …).
- A **cell** is where a row and column meet; it is named by its **cell address** or reference, e.g. **B3** (column B, row 3).
- The **active cell** has a dark border; its address shows in the **Name Box**.
- The **Formula bar** (Input line) shows the contents of the active cell.
- A **range** is a group of cells, written as **A1:C5** (from A1 to C5).
- Sheet tabs at the bottom; add a sheet with **Sheet > Insert Sheet** or the **+** button.

Entering data

Data can be **text (labels)**, **numbers** or **formulas**. Text aligns left, numbers align right by default. Use **AutoFill**: type 1 and 2 in two cells, select both and drag the **fill handle** (small square at the bottom-right) to continue the series. Typing *Monday* and dragging fills the days of the week.

Formulas and operators

A **formula** always starts with **=**. It can use cell references, so the result updates when data changes.

Operator	Meaning	Example
+	Addition	=A1+B1
–	Subtraction	=A1–B1
*	Multiplication	=A1*B1
/	Division	=A1/B1
^	Exponent (power)	=A1^2
%	Percent	=A1*10%

Order of calculation follows **BODMAS**: brackets, then ^, then * and /, then + and –. So `=2+3*4` gives **14**, but `=(2+3)*4` gives **20**.

Functions

Functions are built-in formulas.

Function	Result
`=SUM(B2:B6)`	Total of the range
`=AVERAGE(B2:B6)`	Mean
`=MAX(B2:B6)` / `=MIN(B2:B6)`	Largest / smallest
`=COUNT(B2:B6)`	Number of cells with numbers
`=ROUND(B2,1)`	Rounds to 1 decimal place
`=IF(B2>=33,"Pass","Fail")`	Checks a condition

REPORT CARD

Marks of 5 subjects in B2:F2. Total in G2: `=SUM(B2:F2)`. Percentage in H2: `=G2/500*100`. Result in I2: `=IF(H2>=33,"Pass","Fail")`. Drag G2:I2 down to fill for all students.

Relative and absolute references

- **Relative** (A1): changes when copied. `=B2*C2` copied down becomes `=B3*C3`.
- **Absolute** (A1): does not change when copied – use for a fixed value like a tax rate in E1: `=B2*$E$1`.
- **Mixed** ($A1 or A$1): only one part is fixed. Press **Shift+F4** in Calc to cycle through reference types.

FORGETTING THE = SIGN

Typing `SUM(A1:A5)` without = shows it as text. Every formula must begin with =.

Sorting and filtering

- **Sort**: select data → **Data > Sort** → choose the column and **Ascending/Descending**.
- **Filter**: **Data > AutoFilter** adds drop-down arrows to the header row to show only matching rows.

Charts

Charts show data visually.

1. Select the data (with headings).
2. **Insert > Chart**.
3. Choose a chart type: **Column**, **Bar**, **Pie**, **Line**, **Area**, **XY (Scatter)**.

4. Set data range, data series, titles and legend in the wizard → **Finish**.

Use a **pie chart** for parts of a whole, a **column/bar chart** to compare values, and a **line chart** to show change over time.

COMMON EXAM TRAP

A question may ask the output of `=10+20/2*3`. Do division and multiplication left to right first: 20/2 = 10, × 3 = 30, + 10 = **40**.

Quick recap

- Calc file: .ods; cell address = column letter + row number; range A1:C5.
- Formulas begin with =; BODMAS order; functions SUM, AVERAGE, MAX, MIN, COUNT, IF.
- Relative refs change when copied; absolute (A1) do not.
- Data > Sort, Data > AutoFilter; Insert > Chart (column, bar, pie, line).

Practice questions

1. The extension of a LibreOffice Calc file is
   (a) .odt (b) .ods (c) .odp (d) .xls only
2. The cell at the intersection of column D and row 7 is
   (a) 7D (b) D:7 (c) D7 (d) R7CD
3. What is the result of =2+3*4?
   (a) 20 (b) 14 (c) 24 (d) 9
4. Which of these is an absolute cell reference?
   (a) A1 (b) $A1 (c) A$1 (d) A1
5. Which function finds the largest value in a range?
   (a) =SUM() (b) =COUNT() (c) =MIN() (d) =MAX()
6. Which chart is most suitable to show the share of each item in a total?
   (a) Pie chart (b) Line chart (c) XY (Scatter) chart (d) Area chart
7. The small square at the bottom-right of a selected cell used to copy a series is the
   (a) Name Box (b) Formula bar (c) fill handle (d) sheet tab
8. Differentiate between relative and absolute cell referencing with an example.
9. Marks of a student in five subjects are in cells B2 to F2. Write formulas to find (i) the total in G2 (ii) the average in H2 (iii) the highest mark in I2. Also write the steps to create a column chart of the marks.

Answers: 1. b 2. c 3. b 4. d 5. d 6. a 7. c 8. see notes 9. see notes